

Fractures of the Upper limb General causes • Fall on an outstretched hand FOOSH • Direct blow or trauma • Road traffic accidents • Sports injuries • Twisting or rotational injuries • Occupational injuries • Osteoporosis/pathological bone weakness **General clinical features** • Pain and tenderness • Swelling and bruising • Deformity or abnormal movement • Loss of function and restricted ROM • Crepitus • Shortening or abnormal alignment • Neurovascular deficit in complicated fractures **General complications** • Neurovascular injury • Compartment syndrome • Infection in open fractures • Delayed union/non union • Malunion • Joint stiffness • Post traumatic arthritis • Complex regional pain syndrome --- **1 Fractures of Clavicle and Scapula A Fracture of clavicle Definition** Fracture of the clavicle most commonly involving the middle third **Causes** • Fall directly on the shoulder • Fall on an outstretched hand • Direct blow to clavicle • Road traffic or sports injury **Clinical features** • Pain and tenderness over clavicle • Swelling and deformity • Patient supports the affected arm • Shoulder drooping forwards and downwards • Abnormal mobility/crepitus • Difficulty in shoulder movements • Check distal neurovascular status **Complications** • Malunion • Non union • Neurovascular injury • Skin perforation • Brachial plexus injury • Pneumothorax • Shoulder stiffness **Conservative management** • Arm sling or figure of eight brace • Analgesics • Ice/elevation initially • Early elbow wrist and hand movements • Gradual shoulder ROM after pain subsides **Surgical management** • ORIF with plate and screws • Indications open fracture neurovascular injury significant displacement/shortening threatened skin or selected unstable displaced fractures **B Fracture of scapula Causes** • Usually high energy trauma • Direct blow • RTA • Fall from height • Indirect injury through shoulder **Clinical features** • Severe shoulder/scapular pain • Swelling and tenderness • Painful shoulder movements • Reduced shoulder ROM • Associated chest injuries may be present **Complications** • Associated rib and chest injuries • Pneumothorax • Neurovascular injury • Malunion • Shoulder stiffness • Glenohumeral instability **Management** • Conservative sling analgesia rest followed by early ROM • Surgical ORIF for significantly displaced body/neck fractures intra articular glenoid fractures open fractures or fractures associated with instability --- **2 Fractures of Greater Tuberosity and Neck of Humerus A Greater tuberosity fracture Causes** • Fall on shoulder/FOOSH • Shoulder dislocation • Direct trauma **Clinical features** • Pain and tenderness over greater tuberosity • Swelling • Painful and restricted abduction • Weakness of rotator cuff • Difficulty raising arm **Complications** • Malunion • Shoulder stiffness • Rotator cuff dysfunction • Subacromial impingement • Non union **Management** • Conservative sling analgesics and early passive ROM when appropriate • Surgical ORIF for significantly displaced fragments particularly when displacement causes impingement or affects shoulder mechanics **B Fracture of neck of humerus Causes** • FOOSH • Fall directly on shoulder • Osteoporotic fall in elderly • High energy trauma in young adults **Clinical features** • Pain and swelling around shoulder • Tenderness • Inability to move shoulder • Bruising • Deformity • Check axillary nerve and distal circulation **Complications** • Axillary nerve injury • Vascular injury • Avascular necrosis of humeral head • Malunion/non union • Shoulder stiffness • Rotator cuff dysfunction **Management** • Conservative sling/shoulder immobilizer analgesia and progressive ROM for undisplaced/stable fractures • Surgical ORIF or arthroplasty for displaced unstable comminuted or selected fracture dislocations --- **3 Fracture Shaft of Humerus Definition** Fracture of the humeral diaphysis between the surgical neck and supracondylar region **Causes** • Fall • Direct blow • RTA • Sports injury • Pathological fracture **Clinical features** • Pain and swelling of arm • Deformity • Abnormal mobility and crepitus • Loss of arm function • Radial nerve palsy wrist drop inability to extend wrist/fingers and sensory changes over the dorsum of hand **Complications** • Radial nerve injury • Brachial artery injury • Non union • Malunion • Infection • Shoulder/elbow stiffness **Conservative management** • Coaptation splint or hanging arm cast initially • Functional brace after swelling decreases • Analgesia • Maintain elbow wrist and hand movements • Regular radiographic follow up **Surgical management** • ORIF with plate and screws • Intramedullary nailing • External fixation in selected open/severe injuries • Indications include open fracture vascular injury multiple injuries unacceptable alignment pathological fracture and failure of conservative treatment --- **4 Supracondylar Fracture of Humerus Definition** Fracture of the distal humerus just above the elbow condyles commonly seen in children **Causes** • Fall on outstretched hand • Fall onto flexed elbow • Sports injury **Clinical features** • Severe elbow pain • Swelling and deformity • Loss of elbow movement • Tenderness • Abnormal carrying angle • Assess brachial artery and median radial and ulnar nerves **Complications** • Brachial artery injury • Median/anterior interosseous nerve injury • Compartment syndrome • Volkmann ischaemic contracture • Cubitus varus • Elbow stiffness • Malunion **Conservative management** • Gartland type I above elbow cast/splint with elbow flexed appropriately and neurovascular monitoring • Regular X ray follow up • Avoid excessive elbow flexion because it may compromise circulation **Surgical management** • Closed reduction and percutaneous pinning CRPP for displaced/unstable fractures • Open reduction and internal fixation when closed reduction fails or in selected complicated fractures • Urgent surgery if there is vascular compromise or an open injury --- **5 Side Swipe Injury of Elbow Definition** A side swipe injury is a high energy injury to the elbow produced by a direct blow classically when the elbow is struck by a passing vehicle **Causes** • Direct high energy blow to the elbow • Road traffic accident **Clinical features** • Severe pain and swelling • Open wound commonly present • Deformity around elbow • Restricted elbow movement • Fracture dislocation may occur • Possible vascular and nerve injury **Complications** • Infection • Neurovascular injury • Compartment syndrome

- Elbow stiffness • Non union/malunion • Post traumatic arthritis

Conservative management • Usually limited because these are commonly high energy unstable injuries • Initial splintage • Analgesia • Wound care • Antibiotics and tetanus prophylaxis for open injuries • Neurovascular monitoring

Surgical management • Urgent irrigation and debridement for open injury • Reduction of dislocation • ORIF of associated fractures • External fixation when there is severe soft tissue injury or instability ---

6 Fractures of Radius and Ulna Both bone forearm fracture

Definition Fracture of both the radius and ulna usually involving their shafts

Causes • Direct blow • FOOSH • RTA • Sports injury • Twisting injury

Clinical features • Pain and swelling • Deformity and shortening • Abnormal movement/crepitus • Loss of pronation and supination • Neurovascular assessment is essential

Complications • Compartment syndrome • Malunion • Non union • Loss of pronation/supination • Nerve and vascular injury • Infection • Radioulnar synostosis

Management • Conservative closed reduction and above elbow cast in selected children/undisplaced fractures • Surgical ORIF with plates and screws is commonly required in adults because accurate restoration of radial bow and forearm rotation is important

Chauffeur fracture

Important correction Chauffeur fracture is NOT a fracture of both radius and ulna It is an intra articular fracture of the radial styloid process

Management • Undisplaced splint/cast • Displaced or unstable reduction and fixation commonly ORIF ---

7 Fractures of Capitulum Radial Head and Olecranon

A Capitulum fracture

Causes • FOOSH • Direct trauma • Elbow dislocation

Clinical features • Elbow pain and swelling • Tenderness over lateral elbow • Restricted flexion extension • Mechanical locking may occur

Complications • Non union • Malunion • Elbow stiffness • Post traumatic arthritis

Management • Undisplaced immobilization followed by early ROM • Displaced ORIF with screws excision of small non reconstructable fragments may be considered in selected cases

B Radial head fracture

Causes • FOOSH • Valgus force at elbow • Direct trauma

Clinical features • Lateral elbow pain • Tenderness over radial head • Painful pronation and supination • Restricted elbow movement • Elbow effusion

Complications • Elbow stiffness • Non union • Malunion • Instability • Post traumatic arthritis

Management • Undisplaced sling briefly followed by early movement • Displaced/unstable ORIF radial head replacement may be required for severely comminuted fractures

C Olecranon fracture

Causes • Direct fall on elbow • Triceps contraction • RTA

Clinical features • Posterior elbow pain and swelling • Tenderness • Palpable fracture gap • Inability to actively extend elbow in displaced fractures

Complications • Non union • Elbow stiffness • Loss of extension • Post traumatic arthritis • Ulnar nerve symptoms

Management • Undisplaced and stable above elbow splint/cast with monitoring • Displaced ORIF using tension band wiring or plate fixation ---

8 Monteggia and Galeazzi Fractures

A Monteggia fracture dislocation

Definition Fracture of the proximal or middle ulna associated with dislocation of the radial head at the radiocapitellar joint

Causes • FOOSH • Direct blow • RTA • Hyperpronation injury

Clinical features • Pain and swelling of forearm/elbow • Deformity • Restricted elbow movement • Prominent/displaced radial head • Possible posterior interosseous nerve palsy

Complications • Radial/PIN injury • Malunion • Non union • Elbow stiffness • Chronic radial head dislocation • Compartment syndrome

Management • Children closed reduction and cast immobilization may be successful when stable • Adults/unstable injuries ORIF of ulna followed by reduction and assessment of radial head stability

B Galeazzi fracture dislocation

Definition Fracture of the middle/distal third of the radial shaft with disruption or dislocation of the distal radioulnar joint DRUJ

Causes • FOOSH with hyperpronation • RTA • High energy trauma

Clinical features • Pain and swelling of forearm/wrist • Radial deformity • Prominent ulnar head • Restricted pronation and supination • DRUJ tenderness/instability

Complications • DRUJ instability • Malunion • Non union • Loss of forearm rotation • Chronic wrist pain • Post traumatic arthritis

Management • Children closed reduction and long arm immobilization may be used when stable • Adults ORIF of radius with plate and screws DRUJ stability is assessed after fixation and stabilized if necessary ---

9 Fractures of Wrist and Hand

Complex Smiths Fracture

Smiths fracture **Definition** Smiths fracture is a fracture of the distal radius with volar displacement/angulation of the distal fragment often considered the reverse of a typical Colles fracture

Causes • Fall onto the dorsum of the flexed wrist • Direct blow to the back of the wrist • RTA

Clinical features • Pain and swelling around wrist • Tenderness over distal radius • Volar deformity • Restricted wrist and finger movements • Possible median nerve symptoms • Reduced grip strength

Complications • Malunion • Median nerve compression/carpal tunnel syndrome • Wrist stiffness • Tendon problems • Post traumatic arthritis • Complex regional pain syndrome

Conservative management • Closed reduction • Below elbow or above elbow cast depending on stability • Elevation and analgesics • Regular X rays to detect loss of reduction • Finger and later wrist ROM exercises

Surgical management • ORIF commonly with a volar locking plate for displaced/unstable or intra articular fractures • External fixation may be used in selected severe comminuted injuries

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Dislocations of the Upper Limb

Definition Dislocation is the complete loss of normal articular contact between the articular surfaces of a joint due to trauma or other causes The shoulder is the most commonly dislocated major joint with anterior dislocation being the most common type ---

1 Shoulder Dislocation

Mechanism of injury • Anterior dislocation Fall on an outstretched arm or forceful combination of abduction extension and external rotation • Posterior dislocation Direct blow to the anterior shoulder with the arm adducted and internally rotated • Violent muscle contraction such as during seizures or electrical injury may cause posterior dislocation ---

A Anterior dislocation of

shoulder Clinical features • Severe pain following trauma • Patient supports the affected arm • Arm held in abduction and external rotation • Loss of normal rounded contour of shoulder • Prominent acromion producing a squared off shoulder • Humeral head may be palpable anteriorly • Complete restriction of active and passive movements • Check axillary nerve and distal vascular status **Complications** • Axillary nerve injury • Axillary artery injury • Greater tuberosity fracture • Hill Sachs lesion • Bankart lesion • Rotator cuff injury • Recurrent dislocation • Joint stiffness Axillary nerve injury is particularly important and should be assessed before and after reduction --- **B Posterior dislocation of shoulder Clinical features** • Severe shoulder pain • Arm held in adduction and internal rotation • Inability to externally rotate the arm • Flattening of the anterior shoulder • Prominent posterior humeral head may be palpable • Restricted shoulder movements **Complications** • Reverse Hill Sachs lesion • Fracture of surgical neck or tuberosity • Neurovascular injury • Rotator cuff injury • Recurrent instability • Osteoarthritis Posterior dislocation is uncommon and may be missed clinically particularly when the patient presents with a guarded internally rotated arm --- **Conservative Management of Shoulder Dislocation 1 Reduction** The dislocation is reduced by closed manipulation preferably with adequate analgesia/sedation and muscle relaxation **Kochers manoeuvre** 1 Patient supine 2 Arm adducted to the side 3 Elbow flexed to 90 4 Slowly externally rotate the arm until resistance is felt 5 Elevate/flex the arm while maintaining external rotation 6 Adduct the arm 7 Internally rotate the shoulder to complete reduction Note Forceful Kocher manoeuvre should be avoided because it can cause fractures and soft tissue injury **Hippocrates manoeuvre** 1 Patient lies supine 2 The affected arm is gently grasped 3 Axillary counter traction is applied 4 Longitudinal traction is applied to the limb 5 Gentle manipulation is continued until the humeral head returns to the glenoid The classical Hippocratic technique is now generally not preferred because of the risk of brachial plexus and vascular injury **After reduction** • Confirm reduction clinically and radiologically • Reassess neurovascular status • Apply sling/immobilizer • Analgesics • Gradual ROM exercises • Physiotherapy for rotator cuff and scapular muscle strengthening --- **Surgical Management Indications** • Irreducible dislocation • Recurrent dislocation • Associated fracture • Significant bone loss • Persistent instability • Major labral injury • Failed conservative treatment **Important operations 1 Bankart repair** • Repair of the detached anteroinferior glenoid labrum and capsule • May be performed arthroscopically or by open technique • Common procedure for recurrent anterior instability **2 Putti Platt operation** • A historical/non anatomic stabilization procedure • Involves shortening and tightening the subscapularis and anterior capsule • Reduces external rotation and therefore limits instability • It has largely been replaced by anatomic Bankart repair in modern practice **3 Latarjet procedure** • Coracoid transfer to the anterior glenoid • Used particularly when there is significant glenoid bone loss or recurrent instability --- **2 Recurrent Dislocation of Shoulder Definition** Repeated episodes of shoulder dislocation or subluxation following an initial traumatic dislocation **Causes** • Inadequate healing of capsule/labrum • Bankart lesion • Glenoid bone loss • Hill Sachs lesion • Ligamentous laxity • Inadequate rehabilitation **Clinical features** • Repeated episodes of dislocation • Apprehension during abduction and external rotation • Pain and instability • Reduced confidence in shoulder movements • Positive apprehension test **Management Conservative** • Activity modification • Shoulder immobilization when indicated • Physiotherapy • Rotator cuff strengthening • Scapular stabilization exercises • Proprioceptive training **Surgical** • Arthroscopic/open Bankart repair • Putti Platt procedure • Latarjet procedure • Remplissage or other bone/soft tissue procedures depending on associated lesions Recurrent instability is particularly common in young active patients after traumatic anterior dislocation - -- **3 Dislocation of Elbow Mechanism of injury** • Commonly due to fall on an outstretched hand • Hyperextension and valgus/varus forces • Sports injuries • Road traffic accidents **Clinical features** • Severe pain • Swelling • Obvious deformity • Loss of elbow movement • Prominent olecranon or abnormal relationship of bony landmarks • Neurovascular assessment is essential **Complications** • Brachial artery injury • Median ulnar or radial nerve injury • Fracture associated with dislocation • Elbow stiffness • Chronic instability • Heterotopic ossification • Post traumatic arthritis **Conservative management** 1 Closed reduction under adequate analgesia/sedation 2 Confirm reduction with X ray 3 Short period of immobilization 4 Early active ROM once stability permits 5 Physiotherapy Early mobilization is important because prolonged immobilization increases the risk of elbow stiffness and poor functional outcome **Surgical management** • Open reduction if irreducible • Repair/reconstruction of collateral ligaments in unstable injuries • Fixation of associated fractures • External fixation for severe instability when required --- **4 Hand Injuries A Crush Injuries Definition** Injury caused by compression of the hand between two objects resulting in damage to skin muscles tendons nerves vessels and bones **Clinical features** • Pain and swelling • Bruising • Skin and soft tissue damage • Fracture/dislocation • Loss of movement • Sensory loss • Vascular compromise **Complications** • Compartment syndrome • Tissue necrosis • Infection • Tendon adhesions • Nerve injury • Stiffness • Contracture • Amputation **Management Conservative** • Elevation • Analgesia • Wound dressing • Splintage • Antibiotics when indicated • Early controlled ROM when safe **Surgical** • Thorough wound exploration and debridement • Repair of tendons nerves and vessels • Fracture fixation • Fasciotomy for compartment syndrome • Skin graft/flap coverage • Amputation when tissue is non viable --- **5 Flexor Tendon Injuries Causes** • Sharp cuts • Glass injuries • Crush injuries • Lacerations **Clinical features** • Loss of active finger flexion • Open wound • Altered resting finger position

- Loss of grip • Sensory or vascular injury may coexist **Management Conservative** • Wound care and splintage in selected partial injuries • Early protected rehabilitation after repair **Surgical** • Primary tendon repair for complete lacerations when appropriate • Secondary tendon reconstruction if primary repair is not possible • Postoperative controlled tendon gliding exercises and splinting --- **6 Extensor Tendon Injuries Causes** • Laceration • Crush injury • Direct trauma • Closed tendon rupture **Clinical features** • Loss of active finger extension • Finger deformity depending on level of injury • Weak grip • Open wound in lacerations **Complications** • Adhesions • Extensor lag • Rupture of repair • Joint stiffness • Deformity **Management Conservative** • Splinting in appropriate position for selected injuries • Controlled mobilization **Surgical** • Tendon repair for appropriate complete lacerations • Tendon graft or transfer for chronic injuries • Postoperative splinting and rehabilitation --- **7 Burn Injuries of Hand Definition** Burn injury is tissue damage caused by thermal chemical electrical or other energy sources **Clinical features** • Pain • Erythema • Blistering • Swelling • Loss of skin and soft tissue in deep burns • Reduced ROM • Circumferential burns may compromise circulation **Complications** • Infection • Edema • Skin and tendon damage • Hypertrophic scar • Contracture • Web space contracture • Burn claw deformity • Loss of finger movement • Functional disability **Conservative management** 1 Assess burn depth and extent 2 Clean and dress the wound 3 Elevate the hand to reduce edema 4 Maintain adequate analgesia 5 Splint in anti deformity/intrinsic plus position • Wrist approximately 20 30 extension • MCP joints 70 90 flexion • IP joints extension • Thumb abducted 6 Early supervised ROM when safe 7 Scar management and pressure therapy during healing 8 Physiotherapy/occupational therapy These measures are important because edema scarring and prolonged immobilization can rapidly destroy hand function which humans apparently decided was worth preserving only after discovering fire **Surgical management** • Escharotomy for circumferential burns with vascular compromise • Early excision/debridement of deep burns • Skin grafting for deep partial thickness and full thickness burns • Flap reconstruction when tendons joints or other deep structures are exposed • Scar release and grafting/flap procedures for established contractures

//////////////////// **Fractures of the Lower Limb General causes** • Road traffic accidents • Fall from height or fall on the limb • Direct blow • Twisting injuries • Sports injuries • Osteoporosis/pathological conditions **General clinical features** • Pain and tenderness • Swelling and bruising • Deformity and shortening • Loss of function • Abnormal mobility and crepitus • Restricted joint movements • Neurovascular deficit in complicated injuries **General complications** • Neurovascular injury • Compartment syndrome • Infection in open fractures • Delayed union/non union • Malunion • Joint stiffness • Post traumatic arthritis • Fat embolism particularly with long bone fractures --- **1 FRACTURE NECK OF FEMUR Definition** Fracture of the femoral neck occurring between the femoral head and trochanteric region It is an intracapsular fracture and may compromise the blood supply of the femoral head **Causes** • Fall in elderly patients commonly with osteoporosis • Direct trauma • Road traffic accidents • Fall from height • Pathological weakening of bone **Classification A Anatomical classification** 1 Subcapital immediately below femoral head 2 Transcervical through middle of neck 3 Basicervical at junction of neck and trochanters **B Garden classification** • Type I Incomplete/valgus impacted fracture • Type II Complete but undisplaced fracture • Type III Complete fracture with partial displacement • Type IV Complete fracture with complete displacement **C Pauwels classification** • Type I Fracture angle 30 • Type II 30 50 • Type III 50 and most vertically oriented/unstable **Clinical features** • Pain in groin/hip • Inability to bear weight • Limb appears shortened • Limb lies in external rotation • Tenderness around hip • Pain on hip movement • In impacted fractures the patient may still be able to walk **Complications** • Avascular necrosis of femoral head • Non union • Malunion • Infection • Implant/fixation failure • Thromboembolism • Hip stiffness • Post traumatic osteoarthritis **Conservative management** Reserved mainly for patients who are medically unfit for surgery or selected stable/impacted fractures • Bed rest initially • Analgesics • Skin traction for pain relief when indicated • Prevention of pressure sores and DVT • Chest physiotherapy • Early mobilization with walking aids as tolerated • Regular radiographic follow up AO notes that non operative treatment is generally reserved for medically infirm patients who cannot tolerate surgery **Surgical management Undisplaced/impacted fractures** • Percutaneous cannulated cancellous screws • Femoral neck system in selected cases **Displaced fractures Young patients** • Urgent reduction • Internal fixation with cannulated screws/DHS/FNS depending on fracture pattern **Elderly patients** • Hemiarthroplasty • Total hip arthroplasty in selected active patients Current AO guidance includes cannulated screws femoral neck system hemiarthroplasty and total hip arthroplasty according to fracture pattern and patient factors --- **2 FRACTURES OF TROCHANTERS Definition** Fractures involving the greater and/or lesser trochanter commonly including intertrochanteric fractures **Causes** • Fall in elderly patients • Direct trauma • RTA • Fall from height • Osteoporosis **Clinical features** • Severe hip pain • Inability to bear weight • Swelling and tenderness over hip • Shortening of limb • External rotation • Painful hip movements • Bruising around hip/thigh **Complications** • Blood loss • Malunion with varus deformity • Non union • Limb shortening • Implant failure • DVT and pulmonary embolism • Infection • Hip stiffness **Conservative management** Used only in selected patients who are unsuitable for surgery • Bed rest • Analgesics • Traction for selected cases • Prevention of pressure sores and DVT • Physiotherapy • Gradual mobilization with walking aids **Surgical management** Preferred in most medically fit patients •

Dynamic hip screw DHS • Proximal femoral nail PFN • Cephalomedullary nail • Dynamic condylar screw in selected patterns • Trochanteric stabilization plate for selected unstable fractures AO describes sliding hip screw trochanteric stabilization and intramedullary nailing among operative options for trochanteric fractures --- **3 FRACTURE SHAFT OF**

FEMUR Definition Fracture involving the diaphysis of the femur extending between the lesser trochanter and the distal metaphyseal region **Mechanism of injury** • High energy trauma RTA fall from height • Direct blow • Twisting force • Pathological fracture • Low energy injury in osteoporotic elderly patients **Clinical features** • Severe thigh pain • Swelling • Deformity • Shortening • Abnormal mobility • Crepitus • Inability to bear weight • Significant blood loss may occur • Assess distal neurovascular status **Complications** • Hemorrhagic shock • Fat embolism syndrome • Compartment syndrome • Neurovascular injury • Infection • Delayed union • Non union • Malunion • Knee/hip stiffness **Conservative**

management Used mainly in selected children or patients unsuitable for surgery • Skin/skeletal traction • Manipulation and reduction • Functional bracing in selected cases • Analgesia • Physiotherapy • Gradual weight bearing after union **Surgical management** Preferred in most adult femoral shaft fractures • Intramedullary interlocking nail commonly preferred • ORIF with plate and screws in selected cases • External fixation for severe open fractures/polytrauma • Early mobilization and rehabilitation AO identifies antegrade nailing and bridge plating among established methods for femoral shaft fractures --- **4 SUPRACONDYLAR FRACTURE OF FEMUR Definition** Fracture of the distal femur occurring just above the femoral condyles **Causes** • High energy RTA • Fall from height • Direct trauma • Osteoporosis in elderly • Pathological fracture **Clinical features** • Severe knee/thigh pain • Swelling • Deformity • Shortening • Inability to bear weight • Restricted knee movement • Possible open wound • Neurovascular assessment is essential **Complications** • Popliteal artery injury • Nerve injury • Compartment syndrome • Malunion • Non union • Knee stiffness • Post traumatic arthritis • Infection **Conservative management** For selected undisplaced fractures or patients unsuitable for surgery • Above knee cast/brace • Traction when indicated • Analgesia • Regular X rays • Quadriceps exercises • Gradual knee ROM and weight bearing **Surgical management** • ORIF with locking compression plate • Retrograde intramedullary nail • External fixation in selected open/complex injuries • Bone grafting for selected non unions --- **5 FRACTURES OF THE CONDYLES OF**

FEMUR Definition Fractures involving the medial or lateral femoral condyle which may extend into the knee joint **Causes** • High energy trauma • Direct blow • Fall • Twisting injury • Osteoporosis **Clinical features** • Knee pain • Swelling and haemarthrosis • Tenderness over affected condyle • Restricted knee movement • Inability to bear weight • Deformity in displaced fractures **Complications** • Knee stiffness • Malunion • Non union • Joint incongruity • Post traumatic osteoarthritis • Neurovascular injury • Infection **Conservative management** For undisplaced and stable fractures • Above knee immobilization • Analgesics • Elevation • Regular radiographs • Quadriceps exercises • Gradual ROM and weight bearing **Surgical management** For displaced/intra articular fractures • Open reduction and internal fixation • Lag screws • Buttress/locking plates • Bone grafting when required • Early controlled knee movement after stable fixation --- **6**

FRACTURE PATELLA Definition Fracture of the patella commonly caused by direct trauma or sudden forceful contraction of the quadriceps **Causes** • Direct fall onto knee • Direct blow • RTA • Sudden forceful quadriceps contraction **Types** • Transverse • Vertical • Comminuted • Marginal • Osteochondral **Clinical features** • Anterior knee pain • Swelling and haemarthrosis • Tenderness • Bruising • Inability to extend knee actively in displaced fractures • Palpable gap between fragments **Complications** • Non union • Malunion • Extensor lag • Knee stiffness • Post traumatic arthritis • Quadriceps weakness • Infection after surgery **Conservative management** For undisplaced fractures with intact extensor mechanism • Knee immobilizer/above knee cast • Knee in extension • Analgesia • Quadriceps exercises • Gradual ROM • Progressive weight bearing **Surgical management** Indicated for displaced fractures or disruption of the extensor mechanism • ORIF with tension band wiring • Cannulated screws with tension band • Plate fixation for comminuted fractures • Partial patellectomy for selected non reconstructable inferior/superior pole fragments • Total patellectomy only in exceptional severely comminuted cases --- **7 FRACTURES OF TIBIAL CONDYLES Definition** Fractures involving the proximal tibial articular surface also called tibial plateau fractures **Causes** • RTA • Fall from height • Sports injuries • Valgus/varus stress • Axial loading • Osteoporosis **Clinical features** • Severe knee pain • Swelling • Haemarthrosis • Tenderness over tibial plateau • Inability to bear weight • Restricted knee movements • Deformity • Possible ligament/meniscal injury **Complications** • Compartment syndrome • Neurovascular injury • Knee stiffness • Malunion • Non union • Post traumatic osteoarthritis • Ligament instability • Infection **Conservative management** For undisplaced/minimally displaced stable fractures • Knee immobilization/hinged brace • Ice and elevation • Analgesics • Non weight bearing • Quadriceps and ankle exercises • Gradual ROM • Progressive weight bearing after radiological healing **Surgical management** For displaced unstable or significantly depressed articular fractures • Closed reduction and percutaneous fixation in selected cases • ORIF with plates and screws • Elevation of depressed articular surface • Bone graft/substitute where required • External fixation for severe soft tissue injury/high energy fractures AO includes ORIF with conventional or locking plate techniques among operative methods for proximal tibial fractures --- **8 BOTH BONE FRACTURE OF TIBIA AND**

FIBULA Definition Fracture of both the tibial and fibular shafts usually caused by significant trauma **Causes** • RTA • Direct

trauma • Fall from height • Sports injuries • Motor vehicle accidents • Direct blow • Twisting force • Osteoporosis • Pathological fracture • Low energy injury in elderly patients **Clinical features** • Severe pain • Swelling • Deformity • Shortening • Abnormal mobility • Crepitus • Inability to bear weight • Significant blood loss may occur • Assess distal neurovascular status **Complications** • Hemorrhagic shock • Fat embolism syndrome • Compartment syndrome • Neurovascular injury • Infection • Delayed union • Non union • Malunion • Knee/hip stiffness **Conservative management** Used mainly in selected children or patients unsuitable for surgery • Skin/skeletal traction • Manipulation and reduction • Functional bracing in selected cases • Analgesia • Physiotherapy • Gradual weight bearing after union **Surgical management** Preferred in most adult femoral shaft fractures • Intramedullary interlocking nail commonly preferred • ORIF with plate and screws in selected cases • External fixation for severe open fractures/polytrauma • Early mobilization and rehabilitation AO identifies antegrade nailing and bridge plating among established methods for femoral shaft fractures --- **4 SUPRACONDYLAR FRACTURE OF FEMUR Definition** Fracture of the distal femur occurring just above the femoral condyles **Causes** • High energy RTA • Fall from height • Direct trauma • Osteoporosis in elderly • Pathological fracture **Clinical features** • Severe knee/thigh pain • Swelling • Deformity • Shortening • Inability to bear weight • Restricted knee movement • Possible open wound • Neurovascular assessment is essential **Complications** • Popliteal artery injury • Nerve injury • Compartment syndrome • Malunion • Non union • Knee stiffness • Post traumatic arthritis • Infection **Conservative management** For selected undisplaced fractures or patients unsuitable for surgery • Above knee cast/brace • Traction when indicated • Analgesia • Regular X rays • Quadriceps exercises • Gradual knee ROM and weight bearing **Surgical management** • ORIF with locking compression plate • Retrograde intramedullary nail • External fixation in selected open/complex injuries • Bone grafting for selected non unions --- **5 FRACTURES OF THE CONDYLES OF**

FEMUR Definition Fractures involving the medial or lateral femoral condyle which may extend into the knee joint **Causes** • High energy trauma • Direct blow • Fall • Twisting injury • Osteoporosis **Clinical features** • Knee pain • Swelling and haemarthrosis • Tenderness over affected condyle • Restricted knee movement • Inability to bear weight • Deformity in displaced fractures **Complications** • Knee stiffness • Malunion • Non union • Joint incongruity • Post traumatic osteoarthritis • Neurovascular injury • Infection **Conservative management** For undisplaced and stable fractures • Above knee immobilization • Analgesics • Elevation • Regular radiographs • Quadriceps exercises • Gradual ROM and weight bearing **Surgical management** For displaced/intra articular fractures • Open reduction and internal fixation • Lag screws • Buttress/locking plates • Bone grafting when required • Early controlled knee movement after stable fixation --- **6**

blow • Fall from height • Sports injury • Twisting injury • Crush injury **Mechanism** • Direct bending force • Indirect twisting force • Combination of bending and rotational forces **Clinical features** • Severe pain • Swelling • Deformity and shortening • Abnormal mobility • Crepitus • Inability to bear weight • Open wound may be present • Neurovascular deficit may occur **Complications** • Compartment syndrome • Infection particularly in open fractures • Delayed union • Non union • Malunion • Limb shortening • Neurovascular injury • Knee/ankle stiffness • Chronic osteomyelitis **Conservative management** For selected stable/undisplaced fractures • Closed reduction • Above knee cast initially • Below knee cast after adequate stability/union as appropriate • Elevation and analgesia • Regular X rays • Non weight bearing followed by progressive weight bearing • Physiotherapy **Surgical management** Common indications include displaced/unstable fractures open fractures vascular injury compartment syndrome and failure of conservative treatment • Intramedullary interlocking nail for tibia • ORIF with plate and screws in selected fractures • External fixation for severe open fractures and major soft tissue injury • Bone grafting for selected delayed/non unions • Early rehabilitation after stable fixation --

- **Dupuytren's Fracture** Important examination correction Dupuytren's fracture is not the usual name for a shaft fracture of both tibia and fibula It traditionally refers to an ankle injury involving a proximal fibular fracture with disruption of the distal tibiofibular syndesmosis and associated medial sided ankle injury historically related to Maisonneuve type injury patterns If your university syllabus specifically lists Both bones fracture of tibia and fibula Dupuytren's write the above tibia/fibula fracture answer and add **Dupuytren's fracture** • Usually produced by external rotation and abduction forces at the ankle • Associated with disruption of the distal tibiofibular syndesmosis • May be associated with medial malleolar or deltoid ligament injury • Clinical features include ankle pain swelling tenderness and difficulty bearing weight • Management depends on stability unstable syndesmotomies generally require reduction and fixation //////////////// **Dislocations of Lower Limb Anterior Posterior and Central Dislocation of Hip Mechanism of Injury** • Usually caused by high energy trauma especially road traffic accidents • Posterior dislocation hip flexed and adducted with a force directed along the femur classically when the knee strikes the dashboard • Anterior dislocation forced abduction and external rotation of the hip • Central dislocation severe axial force directed through the femoral head causing it to be driven medially through the acetabulum usually with an acetabular fracture --- **1 Anterior Dislocation of Hip Definition** Displacement of the femoral head anteriorly from the acetabulum **Mechanism** • Forceful abduction external rotation • May occur with extension or flexion depending on the subtype • Commonly occurs in RTA or falls **Types** 1 Superior/anterior pubic 2 Inferior/anterior obturator **Clinical Features** • Severe hip pain • Inability to bear weight • Limb usually lies in abduction and external rotation • Superior type hip is relatively extended • Inferior type hip is flexed and abducted • Femoral head may be palpable anteriorly • Limb may appear shortened • Restricted and painful hip movements • Neurovascular examination is essential **Complications** • Femoral nerve/artery/vein injury • Femoral head fracture • Acetabular fracture • Avascular necrosis of femoral head • Post traumatic osteoarthritis • Recurrent instability • Sciatic nerve injury is less common than in posterior dislocation **Management** • Immediate trauma assessment and neurovascular examination • X ray to confirm dislocation and identify associated fracture • Urgent closed reduction under adequate sedation/anesthesia • Post reduction X ray/CT to confirm concentric reduction and identify associated injuries • Open reduction if irreducible unstable or associated with fractures requiring fixation --- **2 Posterior Dislocation of Hip Definition** Displacement of the femoral head posteriorly from the acetabulum It is the most common type of traumatic hip dislocation **Mechanism** • Hip is flexed adducted and internally rotated • Sudden force is transmitted along the femur • Classical example dashboard injury during an RTA **Clinical Features** • Severe hip pain • Inability to bear weight • Limb is • Flexed • Adducted • Internally rotated • Shortened • Prominent greater trochanter • Femoral head may be palpable posteriorly • Severe restriction of hip movements • Assess sciatic nerve function **Complications** • Sciatic nerve injury • Avascular necrosis of femoral head • Femoral head fracture • Acetabular fracture • Post traumatic osteoarthritis • Recurrent dislocation • Chondral injury • Associated knee injuries **Management Conservative/closed** • Urgent closed reduction under adequate sedation/anesthesia • Allis manoeuvre is commonly used for posterior dislocation • Reassess neurovascular status after reduction • Post reduction X ray and usually CT to detect associated fractures or intra articular fragments • Protected weight bearing and rehabilitation according to associated injuries **Surgical** • Open reduction when • Closed reduction fails • Dislocation is irreducible • Reduction is non concentric • Associated femoral head/neck or acetabular fracture requires fixation • Hip remains unstable after reduction --- **3 Central Dislocation of Hip Definition** Central dislocation is a medial displacement of the femoral head through a fractured acetabulum It is therefore essentially a fracture dislocation of the acetabulum rather than a simple dislocation **Mechanism** • Severe axial loading transmitted through the femoral shaft • Fall from height • RTA • Direct blow to the greater trochanter • Force drives the femoral head medially into/through the acetabular floor **Clinical Features** • Severe hip pain • Inability to bear weight • Marked restriction of hip movements • Swelling around hip • Limb may appear shortened • Abnormal position of femoral head may be difficult to appreciate clinically • Associated pelvic injuries may be present • Neurovascular assessment is essential **Complications** • Acetabular fracture • Pelvic/visceral injury • Blood loss • Sciatic nerve injury • Avascular necrosis • Post traumatic osteoarthritis • Hip stiffness • Heterotopic

ossification • Malunion/non union of acetabular fracture **Management Initial** • ATLS assessment • Analgesia • Neurovascular examination • X ray pelvis/hip • CT scan to define acetabular fracture pattern **Conservative** • Skeletal traction may be used in selected stable minimally displaced injuries • Analgesics • Non weight bearing • Physiotherapy and gradual mobilization **Surgical** • Indicated for displaced acetabular fractures incongruent hip joint or instability • Open reduction and internal fixation ORIF of the acetabulum • Associated femoral head/neck fractures are treated appropriately • Total hip arthroplasty may be required in selected severe injuries or secondary arthritis

////////////////// **Fracture of Cervical Spine Definition** A cervical spine fracture is a break in the continuity of one or more cervical vertebrae with or without associated ligament injury dislocation or spinal cord injury Cervical spine trauma may cause spinal cord compression and neurological deficit including quadriplegia --- **Causes** • Road traffic accidents • Fall from height • Diving into shallow water • Sports injuries • Direct blow to the head or neck • Hyperflexion or hyperextension injuries • Axial loading of the cervical spine • Crush injuries --- **Mechanism of Injury** 1 Hyperflexion • Sudden forward bending of the neck • May cause vertebral body compression flexion distraction injury or fracture dislocation • Can produce spinal cord injury 2 Hyperextension • Sudden backward bending of the neck • Common in road traffic accidents and falls • May cause extension injuries and spinal cord compression 3 Axial compression • Force transmitted vertically through the head • Common in diving accidents and falls onto the head • May produce burst type fractures 4 Rotation • Rotational force may produce facet injuries and fracture dislocations 5 Combined mechanism • Flexion extension rotation and axial loading may occur together in high energy trauma --- **Clinical Features Local features** • Severe neck pain • Cervical muscle spasm • Tenderness over cervical spine • Swelling • Restricted neck movement • Deformity or abnormal neck position **Neurological features** Depending on the level and severity of spinal cord injury • Weakness or paralysis of limbs • Loss of sensation • Tingling/numbness • Loss of reflexes initially • Respiratory difficulty • Bladder and bowel dysfunction **Quadriplegia** Quadriplegia/tetraplegia is paralysis or significant impairment of motor and/or sensory function in the four limbs and trunk due to cervical spinal cord injury High cervical injuries may cause • Paralysis of all four limbs • Loss of sensation • Respiratory muscle paralysis • Loss of bladder and bowel control • Autonomic dysfunction The neurological level and completeness of spinal cord injury should be assessed using the ASIA/ISNCSCI neurological examination --- **Management of Cervical Spine Fracture 1 Initial Management A Immobilization** Until cervical spine injury is excluded maintain spinal precautions • Keep patient supine • Maintain manual in line stabilization during handling and airway procedures • Apply a properly fitted rigid cervical collar when indicated • Avoid unnecessary neck movements • Log roll the patient while maintaining spinal alignment • Assess airway breathing and circulation • Perform repeated neurological examinations Rigid cervical immobilization is particularly important when there is neurological deficit or suspected significant cervical injury **B Investigation** • Cervical spine CT scan is commonly used to identify fractures and alignment abnormalities • MRI is useful for spinal cord disc ligament and soft tissue abnormalities particularly when neurological deficit is present --- **2 Management for Stabilization Conservative management** Used for stable fractures without significant displacement or neurological compromise • Rigid cervical collar • Cervical orthosis/brace • Analgesics • Activity restriction • Regular radiographs/CT as required • Physiotherapy after adequate healing Stable cervical compression fractures may be treated with a cervical collar for several weeks **Halo immobilization** • Halo vest may provide rigid external immobilization • Used in selected stable or reducible cervical injuries • Requires careful monitoring for complications **Surgical stabilization** Indications • Unstable fracture • Significant displacement or deformity • Fracture dislocation • Progressive neurological deficit • Spinal cord compression • Failure of conservative treatment • Significant ligamentous disruption Procedures • Anterior cervical decompression and fusion • Posterior cervical fixation and fusion • Combined anterior and posterior stabilization when required • Removal of compressive bone/disc fragments when indicated The choice of surgical approach depends on the fracture pattern instability neurological status and site of spinal cord compression --- **3 Management of Quadriplegia Immediate management** 1 Maintain cervical immobilization and spinal alignment 2 Maintain a patent airway 3 Provide oxygen and ventilatory support when required 4 Monitor respiratory function continuously 5 Maintain adequate circulation and blood pressure 6 Perform complete neurological assessment 7 CT/MRI to identify spinal cord compression 8 Urgent spine/neurosurgical consultation 9 Surgical decompression and stabilization when indicated 10 Prevent secondary complications **Prevention of complications** • Regular repositioning • Pressure sore prevention • Chest physiotherapy • Assisted coughing and respiratory care • DVT prophylaxis • Bladder and bowel management • Prevention of contractures • Passive ROM exercises • Physiotherapy and occupational therapy • Psychological and social rehabilitation Patients with cervical spinal cord injury may require prolonged multidisciplinary rehabilitation --- **4 Clay Shovelers Fracture Definition** A Clay shovelers fracture is an avulsion fracture of the spinous process most commonly involving C7 but it may involve C6 to T3 **Mechanism** • Direct blow to the posterior neck • Sudden forceful flexion • Sudden muscular/ligamentous traction on the spinous process • Sports injuries • RTA **Clinical features** • Sudden localized neck pain • Pain between the shoulder blades • Local tenderness • Swelling • Restricted cervical movements • Usually no neurological deficit **Complications** • Chronic neck pain • Neck stiffness • Rare

neurological injury • Non union may occasionally cause persistent pain **Management Conservative** • Rest • Analgesics/NSAIDs • Hard cervical collar for comfort • Gradual return to activity • Physiotherapy after pain subsides Most isolated clay shovelers fractures are stable injuries and heal with conservative treatment **Surgical** • Rarely required • Excision of the painful non united fragment may be considered in persistent symptomatic cases --- **5 Hangmans**

Fracture Definition A Hangmans fracture is a traumatic fracture through the bilateral pars interarticularis/neural arch of C2 often associated with traumatic C2 C3 spondylolisthesis **Mechanism** • Hyperextension • Axial loading • Often associated with high energy trauma • Examples RTA fall from height diving injury **Clinical features** • Severe neck pain • Neck stiffness • Tenderness • Restricted cervical movements • Possible C2 C3 deformity • Neurological deficit is relatively uncommon in typical injuries because the spinal canal may become widened **Complications** • C2 C3 instability • Non union • Malunion • Chronic neck pain • Neurological injury in severe displacement • Vertebral artery injury • Post traumatic deformity --

- **Classification of Hangmans Fracture Levine Edwards classification** **Type I** • Hyperextension and axial compression • Minimal displacement • Stable injury **Type II** • Hyperextension and axial loading followed by flexion • Significant displacement and angulation • Unstable **Type IIA** • Flexion distraction injury • Significant angulation • Usually little horizontal displacement **Type III** • Type I injury associated with bilateral C2 C3 facet dislocation • Highly unstable --

- **Management of Hangmans Fracture Conservative** For stable/minimally displaced injuries • Rigid cervical collar • Analgesics • Regular radiological follow up • Halo immobilization for selected unstable but reducible injuries • Gradual mobilization after healing Type I injuries can commonly be treated with a rigid cervical collar **Surgical Indications** • Significant displacement • Severe angulation • Failure of closed reduction • C2 C3 facet dislocation • Severe instability • Neurological compression • Failure of conservative treatment **Procedures** • Posterior C2 C3 fixation and fusion • Anterior C2 C3 fusion in selected cases • Combined fixation for complex injuries //////////////// **Fractures of Thoracic and Lumbar Spine Definition** A thoracic or lumbar spine fracture is a break in one or more vertebral structures of the thoracic or lumbar spine with or without associated ligament spinal cord or nerve root injury Thoracolumbar fractures commonly occur around the T11 L2 region which represents the transition between the relatively rigid thoracic spine and mobile lumbar spine --- **1 Mechanism of Injury** Thoracolumbar fractures commonly result from 1 Flexion compression • Sudden forward flexion of the spine • Produces compression/wedge fractures • Common after falls and road traffic accidents 2 Axial compression • Vertical force transmitted through the spine • Produces burst fractures with compression of the vertebral body and possible retropulsion into the spinal canal 3 Flexion distraction • Excessive forward bending with distraction of the posterior elements • Produces Chance fracture • Classically associated with seat belt injuries 4 Hyperextension • Forced backward bending • May produce anterior column injury and extension type fractures 5 Rotation/shear • High energy rotational or translational forces • May produce fracture dislocation and severe spinal instability --- **2 Common Thoracic and Lumbar Fractures A Compression/Wedge Fracture Definition** Collapse of the anterior portion of the vertebral body producing a wedge shaped vertebra **Causes** • Osteoporosis • Fall • RTA • Axial loading **Clinical features** • Localized back pain • Tenderness • Muscle spasm • Increased pain with movement • Kyphotic deformity may develop • Usually no neurological deficit in a simple stable fracture **Management Conservative** • Analgesics • Short period of rest • Thoracolumbosacral orthosis TLSO in selected cases • Early mobilization • Physiotherapy • Osteoporosis management where appropriate **Surgical** • Rare in stable fractures • Consider vertebral augmentation or fixation in selected patients with persistent severe pain progressive deformity or instability --- **3 Burst Fracture Definition** A burst fracture occurs when axial compression causes the vertebral body to fracture in multiple directions potentially pushing bone fragments into the spinal canal **Mechanism** • Severe axial loading • Fall from height landing on feet • RTA • Diving injury **Clinical features** • Severe back pain • Local tenderness • Muscle spasm • Restricted spinal movements • Possible kyphosis • Neurological deficit may occur due to canal compromise **Complications** • Spinal cord/conus injury • Nerve root compression • Progressive kyphosis • Chronic pain • Neurological deterioration • Spinal instability **Conservative management** For stable fractures without neurological deficit • Analgesics • TLSO brace in selected patients • Early mobilization • Physiotherapy • Serial radiographs **Surgical management** Indications include • Significant instability • Progressive deformity • Neurological deficit with compression • Significant canal compromise in appropriate clinical settings • Failure of conservative treatment **Procedures** • Posterior pedicle screw fixation • Decompression when indicated • Vertebral reconstruction/corpectomy in selected severe injuries • Fusion when instability requires it AO describes non operative treatment for selected stable thoracolumbar injuries and operative stabilization for unstable patterns surgeryreferenceaofoundationorg --- **4 Chance Fracture Definition** A Chance fracture is a flexion distraction injury extending horizontally through the vertebral body and posterior elements **Mechanism** • Sudden flexion distraction • Classically caused by a lap seat belt during RTA **Clinical features** • Severe back pain • Local tenderness • Paraspinal muscle spasm • Possible abdominal bruising/seat belt sign • Neurological deficit may occur **Complications** • Spinal instability • Kyphotic deformity • Chronic back pain • Neurological injury • Associated abdominal/visceral injury **Management Conservative** • Selected stable bony injuries with intact posterior ligamentous complex • TLSO brace • Analgesia • Regular radiological

follow up • Physiotherapy **Surgical** • Indicated for ligamentous disruption significant displacement instability progressive deformity or neurological compromise • Posterior pedicle screw fixation and fusion is commonly used --- **5 Fracture Dislocation of Thoracolumbar Spine Mechanism** • High energy trauma • Severe flexion extension rotation or shear forces • RTA • Fall from height **Clinical features** • Severe back pain • Obvious deformity • Severe muscle spasm • Inability to move • Neurological deficit may be present **Complications** • Spinal cord/conus injury • Cauda equina syndrome • Permanent neurological deficit • Spinal instability • Deformity • Chronic pain **Management Initial** • Full spinal immobilization • ABC assessment • Neurological examination • CT/MRI as indicated **Surgical** • Usually requires reduction and internal fixation with pedicle screws and rods often with fusion • Decompression if neural structures are compressed • Postoperative rehabilitation These injuries are generally considered unstable and require specialist spinal management surgeryreferenceaofoundationorg --- **6 General Clinical Features of Thoracic and Lumbar Fractures Local features** • Severe localized back pain • Vertebral tenderness • Paraspinal muscle spasm • Swelling • Restricted spinal movements • Kyphotic deformity • Bruising **Neurological features** Depending on the level and severity • Lower limb weakness/paralysis • Numbness or paraesthesia • Altered reflexes • Sensory loss • Bladder and bowel dysfunction • Sexual dysfunction • Cauda equina/conus symptoms in lower thoracolumbar injuries **Associated injuries** High energy trauma may be associated with • Rib fractures • Chest injury • Abdominal injury • Pelvic fractures • Long bone fractures --- **7 General Management A Initial Management** 1 Spinal precautions and immobilization 2 ABC assessment 3 Maintain adequate oxygenation and circulation 4 Neurological examination 5 Analgesia 6 CT scan for bony injury 7 MRI when neurological deficit ligamentous injury or spinal cord compression is suspected --- **B Conservative Management** Indicated mainly for stable fractures without neurological deficit • Bed rest only for a short period when necessary • Analgesics • TLSO brace where appropriate • Early mobilization • Physiotherapy • Back and abdominal muscle strengthening • Gradual return to activities • Regular radiological follow up Prolonged bed rest should generally be avoided because it increases complications such as DVT pressure sores and deconditioning --- **C Surgical Management Indications** • Unstable fracture • Progressive kyphosis • Significant displacement • Neurological deficit due to compression • Spinal cord/conus/cauda equina compression • Disruption of posterior ligamentous complex • Fracture dislocation • Failure of conservative treatment **Procedures** • Posterior pedicle screw fixation • Posterior spinal fusion • Decompression/laminectomy when indicated • Anterior decompression and fusion in selected cases • Combined anterior and posterior fixation for severe injuries --- **8 Fracture of Coccyx Definition** A coccygeal fracture is a break in one or more segments of the coccyx usually caused by a direct fall onto the buttocks **Causes** • Fall directly on the buttocks • Fall from height • Direct blow • Childbirth • Occasionally repetitive trauma **Mechanism** • Direct axial force on the coccyx • Sudden backward bending of the coccyx • Direct impact while sitting/falling **Clinical Features** • Pain over coccyx coccydynia • Local tenderness • Pain while sitting • Pain during rising from a chair • Pain during defecation • Pain during prolonged sitting • Bruising/swelling may be present • Pain on direct palpation **Complications** • Chronic coccydynia • Non union • Malunion • Painful sitting • Difficulty with defecation • Rare rectal injury --- **Conservative Management** Most coccygeal fractures are treated conservatively 1 Analgesics/NSAIDs 2 Avoid prolonged sitting 3 Use a coccygeal cushion or wedge cushion 4 Sit leaning slightly forward to reduce pressure on the coccyx 5 Ice during the acute phase 6 Stool softeners if painful defecation is present 7 Gradual return to normal activity 8 Physiotherapy for persistent symptoms **Surgical Management** Surgery is rarely required • Considered for severe persistent pain despite prolonged conservative treatment • Coccygectomy may be performed in carefully selected patients with chronic refractory coccydynia //////////////////////////////////// **FRACTURE OF RIB CAGE Definition** A fracture of the rib cage is a break in the continuity of one or more ribs or the sternum usually caused by direct or indirect trauma Severe chest trauma may be associated with pneumothorax haemothorax pulmonary contusion or flail chest --- **1 FRACTURE OF RIBS Mechanism of Injury** • Direct trauma A blow to the chest causes the rib to bend and fracture at the site of impact • Indirect trauma Compression of the chest from the sides may fracture ribs at their point of maximum curvature • RTA • Fall • Sports injury • Crush injury • Severe coughing may rarely cause stress fractures **Clinical Features** • Severe localized chest pain • Pain aggravated by • Deep inspiration • Coughing • Sneezing • Movement • Local tenderness over the fractured rib • Swelling and bruising • Painful and shallow breathing • Crepitus may be present • Reduced chest expansion on the affected side • Respiratory distress in severe injuries **Complications** • Pneumothorax • Haemothorax • Pulmonary contusion • Pneumonia • Atelectasis • Respiratory failure • Flail chest • Injury to underlying organs **Management Conservative** • Assess airway breathing and circulation • Analgesics including multimodal analgesia when required • Ice application during the acute phase • Encourage deep breathing and coughing exercises • Incentive spirometry where appropriate • Chest physiotherapy • Early mobilization • Treat associated pneumothorax/haemothorax if present Important Routine tight chest binding is avoided because it restricts ventilation and may increase pulmonary complications **Surgical** Usually not required for uncomplicated rib fractures Indications may include • Severe chest wall instability • Flail chest in selected patients • Multiple severely displaced fractures • Persistent respiratory compromise • Failure of conservative treatment Surgical stabilization of rib fractures SSRF may be performed using plates and screws in selected patients --- **2 FRACTURE OF**

STERNUM Definition A fracture of the sternum is a break in the continuity of the breastbone commonly occurring after high energy blunt trauma **Mechanism of Injury** • Direct blow to anterior chest • RTA particularly impact against steering wheel or restraint system • Fall onto the chest • Compression injury • Severe chest trauma **Clinical Features** • Severe anterior chest pain • Localized tenderness • Swelling and bruising • Pain on deep inspiration • Pain on coughing • Crepitus or palpable deformity • Reduced chest movement • Associated respiratory distress if there are intrathoracic injuries **Complications** • Myocardial injury • Arrhythmias • Pulmonary contusion • Pneumothorax • Haemothorax • Rib fractures • Rare vascular injury • Respiratory complications Because sternal fractures can indicate significant blunt chest trauma associated cardiac and thoracic injuries must be assessed **Management Conservative** For uncomplicated stable fractures • Analgesics • Ice application • Rest initially • Deep breathing exercises • Coughing exercises • Chest physiotherapy • Early mobilization • Monitor for associated cardiopulmonary injury **Surgical** Rarely required Indications • Severe displacement • Unstable fracture • Persistent severe pain • Non union • Associated chest wall instability • Failure of conservative treatment ORIF with plates and screws may be performed when surgical stabilization is indicated --- **3**

FRACTURE OF PELVIS Definition A pelvic fracture is a break in one or more bones forming the pelvic ring including the ilium ischium pubis or sacrum Pelvic fractures range from stable injuries to life threatening unstable pelvic ring injuries **Mechanism of Injury** 1 High energy trauma • RTA • Fall from height • Crush injury • Industrial accidents 2 Low energy trauma • Falls in elderly osteoporotic patients **Mechanisms of pelvic ring injury** • AP compression pelvis opens like a book • Lateral compression inward compression of the pelvic ring • Vertical shear upward displacement of one side of the pelvis • Combined mechanisms may occur --- **Clinical Features Local** • Severe pelvic/hip pain • Inability to stand or walk • Pelvic tenderness • Swelling and bruising • Limb length discrepancy • Pelvic deformity • Pain on movement **Associated features** • Shock due to major blood loss • Abdominal injury • Urethral/bladder injury • Rectal injury • Nerve injury • Lower limb vascular injury **Complications** • Massive haemorrhage and hypovolaemic shock • Pelvic organ injury • Bladder/urethral injury • Rectal injury • Neurovascular injury • DVT and pulmonary embolism • Infection • Malunion • Chronic pelvic pain • Gait abnormality • Sexual dysfunction --- **Management of Pelvic Fracture A Initial Management** Pelvic fractures are managed initially according to trauma principles 1 Airway and breathing assessment 2 Circulation assessment 3 Control external haemorrhage 4 Avoid repeated manipulation of the pelvis 5 Apply a pelvic binder when an unstable pelvic ring injury with suspected haemorrhage is present 6 Establish IV access 7 Fluid and blood product resuscitation as required 8 Assess for abdominal urinary neurological and vascular injuries 9 Imaging with X ray/CT according to the patients condition A pelvic binder is used as an early temporary measure for suspected unstable pelvic ring injury and should be positioned appropriately around the greater trochanters --- **B Conservative Management** Used mainly for stable minimally displaced pelvic fractures • Bed rest initially when required • Analgesics • Pelvic support where appropriate • DVT prophylaxis • Early mobilization • Walking aids • Physiotherapy • Progressive weight bearing • Regular radiological follow up --- **C Surgical Management** Indications • Unstable pelvic ring fracture • Significant displacement • Open pelvic fracture • Persistent haemodynamic instability • Failure of conservative management • Associated organ/vascular injury Methods • External fixation • Pelvic packing for uncontrolled pelvic haemorrhage in appropriate settings • Angioembolization for arterial bleeding • ORIF with plates and screws • Percutaneous screw fixation in selected fractures

////////// **INFLAMMATION AND REPAIR Introduction to Soft Tissue Healing Definition of Inflammation** Inflammation is a protective response of living vascularized tissue to injury infection or irritation which removes the injurious stimulus and initiates the process of healing and repair --- **1 Causes of Inflammation** • Trauma and mechanical injury • Burns • Infection • Chemical irritants • Foreign bodies • Tissue necrosis • Ischaemia • Immune reactions --- **2 Types of Inflammation A Acute inflammation** • Rapid onset • Usually lasts for a short duration • Predominantly involves neutrophils • Characterized by vascular changes exudation of fluid and inflammatory cells **B Chronic inflammation** • Develops gradually or persists for a longer duration • Predominantly involves lymphocytes macrophages and plasma cells • Associated with tissue destruction and fibrosis --- **3 Cardinal Signs of Inflammation** 1 Rubor redness 2 Calor heat 3 Tumor swelling 4 Dolor pain 5 Functio laesa loss of function --- **4 Basic Process of Inflammation Vascular response** • Brief vasoconstriction • Vasodilation • Increased blood flow • Increased vascular permeability • Formation of inflammatory exudate • Stasis of blood flow **Cellular response** • Margination of leukocytes • Rolling • Adhesion • Transmigration/diapedesis • Chemotaxis • Phagocytosis • Destruction/removal of microorganisms and damaged tissue --- **5 Definition of Repair** Repair is the restoration of tissue continuity following injury by replacement of damaged tissue with new tissue Repair occurs through two major processes 1 Regeneration 2 Scar formation **Regeneration** Replacement of damaged cells by cells of the same type restoring normal tissue structure and function **Scar formation** Replacement of damaged tissue by connective tissue resulting in fibrosis and scar formation --- **6 Soft Tissue Healing** Soft tissue healing is the process by which injured skin muscle tendon ligament and other soft tissues restore their structural integrity and function following injury Healing occurs through overlapping phases **Phase 1 Haemostasis** Immediately after injury • Vasoconstriction • Platelet aggregation • Formation of blood clot • Fibrin clot formation • Provides temporary wound stability • Releases growth factors that initiate

healing --- **Phase 2 Inflammatory Phase** Approximately first few days • Vasodilation and increased vascular permeability • Neutrophils migrate to the wound • Macrophages subsequently predominate • Removal of bacteria dead cells and debris • Release of cytokines and growth factors • Prepares the wound for tissue formation --- **Phase 3 Proliferative Phase** Usually begins within a few days and continues for several weeks Main events • Fibroblast proliferation • Collagen synthesis • Formation of granulation tissue • Angiogenesis • Re epithelialization • Wound contraction Initially the wound contains predominantly type III collagen --- **Phase 4 Remodeling/Maturation** Begins after the proliferative phase and may continue for months to years • Collagen is reorganized • Type III collagen is gradually replaced by stronger type I collagen • Collagen cross linking increases • Vascularity decreases • Scar becomes paler and flatter • Tensile strength progressively increases but does not normally return completely to that of uninjured tissue The inflammatory proliferative and remodeling phases overlap rather than occurring as completely separate stages --- **7 Factors Affecting Soft Tissue Healing Local factors** • Poor blood supply • Infection • Tissue damage • Foreign body • Excessive tension • Repeated trauma • Haematoma • Excessive movement **Systemic factors** • Age • Poor nutrition • Diabetes mellitus • Smoking • Anaemia • Immunosuppression • Certain medications particularly corticosteroids --- **8 Types of Wound Healing Primary intention** • Wound edges are closely approximated • Minimal tissue loss • Less granulation tissue • Small scar • Example clean surgical incision **Secondary intention** • Wound edges cannot be approximated • Greater tissue loss • Extensive granulation tissue • More wound contraction • Larger scar **Tertiary intention** • Wound is initially left open because of contamination/infection and is closed later after the wound is adequately clean ////////////// **1 Sprain Definition** Injury to a ligament caused by excessive stretching or tearing usually due to sudden twisting or abnormal joint movement **Features** Pain swelling tenderness and joint instability **2 Strain Definition** Injury to a muscle or tendon caused by overstretching or excessive force **Features** Pain muscle spasm weakness and swelling **3 Contusion Definition** Injury to soft tissue caused by a direct blunt blow producing damage to underlying tissues without breaking the skin ////////////// **2 Meniscal Injuries of Knee Mechanism** • Twisting of a weight bearing knee • Sudden change of direction • Deep squatting • Direct trauma • Degenerative injury in older individuals **Clinical Features** • Joint line pain and tenderness • Swelling often developing after activity • Locking or catching • Clicking • Restricted knee movement • Pain on squatting • Positive McMurray test and/or Apley compression test --- **3 Cruciate Ligament Injuries of Knee Mechanism ACL injury** • Sudden deceleration • Pivoting/change of direction • Hyperextension • Valgus force with rotation **PCL injury** • Dashboard injury • Fall onto a flexed knee • Hyperflexion • Posteriorly directed force on the tibia **Clinical Features** • Pain and swelling • Acute haemarthrosis particularly with ACL injury • Feeling of instability or giving way • Restricted ROM • ACL Positive Lachman and anterior drawer tests • PCL Positive posterior drawer test and posterior sag sign --- **4 Medial and Lateral Collateral Ligament Injuries Mechanism MCL** • Valgus stress applied to the knee • Commonly associated with contact sports **LCL** • Varus stress applied to the knee • May occur with high energy trauma **Clinical Features** • Pain and tenderness over the injured ligament • Local swelling • Pain on stress testing • Medial instability with valgus stress in MCL injury • Lateral instability with varus stress in LCL injury • Difficulty in weight bearing --- **5 Lateral Ligament Injury of Ankle Mechanism** Most commonly caused by inversion injury of a plantar flexed ankle Commonly injured ligaments • Anterior talofibular ligament ATFL • Calcaneofibular ligament CFL • Posterior talofibular ligament PTFL **Clinical Features** • Pain around lateral ankle • Swelling • Bruising • Tenderness over lateral ligaments • Difficulty in weight bearing • Reduced ankle ROM • Instability in severe injury • Positive anterior drawer/talar tilt tests when appropriate --- **6 Wrist Sprain Mechanism** • Fall on an outstretched hand FOOSH • Sudden hyperextension or hyperflexion • Twisting injury • Direct trauma **Clinical Features** • Wrist pain • Swelling • Tenderness • Painful and restricted wrist movement • Reduced grip strength • Bruising • Pain during loading of the wrist ////////////// **Strains Contusions and Tendon Ruptures 1 Strains Definition** A strain is an injury involving a muscle or musculotendinous unit caused by excessive stretching or forceful contraction **A Quadriceps strain Mechanism** Sudden forceful contraction sprinting kicking or jumping **Clinical features** • Anterior thigh pain • Tenderness • Swelling/bruising • Pain on resisted knee extension • Weakness of quadriceps **B Hamstring strain Mechanism** Sudden sprinting acceleration or excessive hip flexion with knee extension **Clinical features** • Posterior thigh pain • Tenderness • Bruising/swelling • Pain on resisted knee flexion • Reduced running ability **C Calf strain Mechanism** Sudden push off jumping or running **Clinical features** • Calf pain • Tenderness • Swelling • Pain on resisted plantar flexion • Difficulty walking **D Biceps strain Mechanism** Sudden forceful elbow flexion or lifting **Clinical features** • Anterior arm pain • Tenderness • Pain on resisted elbow flexion/supination • Weakness **E Triceps strain Mechanism** Forceful elbow extension or throwing activity **Clinical features** • Posterior arm pain • Tenderness • Pain on resisted elbow extension • Weakness and swelling -- **2 Contusions Definition** A contusion is a blunt trauma injury causing damage to muscle and small blood vessels without disruption of the skin **A Quadriceps contusion** • Usually caused by direct blow to the thigh • Pain swelling and bruising • Reduced knee flexion • Muscle stiffness • Severe cases may develop myositis ossificans **B Gluteal contusion** • Direct fall or blow to buttock • Pain and tenderness • Swelling and bruising • Pain while sitting or walking **C Calf contusion** • Direct blow to calf • Pain and swelling • Bruising • Tenderness • Painful walking and plantar flexion **D Deltoid contusion** • Direct blow

to shoulder • Shoulder pain and tenderness • Swelling/bruising • Painful shoulder movement • Reduced abduction **General management** Protection/rest initially ice compression and elevation where appropriate followed by gradual ROM and strengthening --- **3 Tendon Ruptures A Achilles Tendon Rupture Mechanism** • Sudden forceful plantar flexion • Jumping or sprinting • Sudden eccentric loading of calf muscles **Clinical features** • Sudden severe pain in posterior ankle • Feeling of being kicked in the calf • Difficulty walking • Weak/absent plantar flexion • Palpable gap above calcaneus • Positive Thompson test **Management** • Immobilization in plantar flexion for selected patients • Functional rehabilitation • Surgical tendon repair in selected active patients complete ruptures and appropriate cases --- **B Rotator Cuff Tendon Rupture Mechanism** • Fall on an outstretched arm • Forceful shoulder movement • Repetitive overhead activity • Degenerative tendon changes **Clinical features** • Shoulder pain • Painful arc • Weakness of shoulder elevation • Difficulty with overhead activities • Night pain • Positive rotator cuff tests **Management** • Rest/activity modification • Analgesics • Physiotherapy ROM and progressive rotator cuff/scapular strengthening • Surgical repair for selected acute traumatic tears significant weakness or failure of conservative treatment //////////////// **AMPUTATIONS Definition and Levels of Amputation Definition** Amputation is the partial or complete removal of a limb or part of a limb through a surgical procedure or as a result of trauma The main aims of amputation are to remove non viable or diseased tissue control infection or pain and create a functional residual limb suitable for rehabilitation and prosthetic fitting --- **1 Levels of Upper Limb Amputation A Finger and Hand** 1 Distal phalanx amputation 2 Middle phalanx amputation 3 Proximal phalanx amputation 4 Metacarpal amputation 5 Ray amputation removal of a digit with its corresponding metacarpal 6 Transmetacarpal amputation through the metacarpals **B Wrist and Forearm** 7 Wrist disarticulation separation through the wrist joint 8 Transradial amputation through the forearm below the elbow **C Elbow and Arm** 9 Elbow disarticulation through the elbow joint 10 Transhumeral amputation through the arm above the elbow **D Shoulder** 11 Shoulder disarticulation removal of the upper limb through the shoulder joint 12 Forequarter amputation removal of the upper limb together with part of the shoulder girdle --- **2 Levels of Lower Limb Amputation A Foot and Ankle** 1 Toe amputation 2 Ray amputation 3 Transmetatarsal amputation 4 Lisfranc amputation through the tarsometatarsal joints 5 Chopart amputation through the transverse tarsal joint 6 Symes amputation ankle disarticulation with preservation of the heel pad **B Leg** 7 Transtibial amputation through the tibia and fibula below the knee **C Knee** 8 Knee disarticulation amputation through the knee joint **D Thigh and Hip** 9 Transfemoral amputation through the femur above the knee 10 Hip disarticulation removal of the limb through the hip joint 11 Hemipelvectomy removal of the lower limb with part of the pelvis --- **3 Indications for Amputation** • Severe traumatic injury • Peripheral vascular disease with irreversible tissue loss • Uncontrolled infection/gangrene • Malignant tumour • Severe congenital deformity • Non functional painful limb • Severe burns or crush injury • Irreversible ischemia --- **4 Principles of Selecting the Level** The level should be selected to • Remove all non viable or diseased tissue • Preserve maximum useful limb length • Preserve viable skin and soft tissue coverage • Maintain adequate blood supply • Preserve important joints whenever possible • Produce a stable painless and functional residual limb • Allow effective prosthetic fitting //////////////// **Traumatic Spinal Cord Injuries Definition** Traumatic spinal cord injury SCI is an acute injury to the spinal cord caused by an external mechanical force resulting in temporary or permanent impairment of motor sensory and autonomic functions below the level of injury --- **Causes** • Road traffic accidents • Falls from height • Sports injuries especially diving • Direct blow to the spine • Gunshot or penetrating injuries • Crush injuries • Hyperflexion or hyperextension injuries • Fracture dislocation of the vertebral column --- **Clinical Features** Clinical presentation depends mainly on the level severity and completeness of spinal cord injury **1 Motor disturbances** • Muscle weakness or paralysis below the level of injury • Paraplegia paralysis of both lower limbs • Quadriplegia/tetraplegia paralysis of all four limbs following cervical cord injury • Loss of voluntary movements • Muscle weakness may be flaccid initially because of spinal shock • Later spasticity and hyperreflexia may develop in upper motor neuron lesions --- **2 Sensory disturbances** • Loss or reduction of sensation below the level of injury • Loss of • Pain • Temperature • Touch • Pressure • Vibration • Proprioception • Paraesthesia or numbness • A clearly defined sensory level may be present --- **3 Reflex changes Spinal shock** Immediately after a significant acute SCI • Flaccid paralysis • Loss of deep tendon reflexes • Loss of superficial reflexes • Absent bulbocavernosus reflex may occur • Areflexia below the level of injury As spinal shock resolves reflexes may return and become exaggerated in upper motor neuron injuries --- **4 Autonomic disturbances Bladder dysfunction** • Urinary retention • Urinary incontinence • Neurogenic bladder **Bowel dysfunction** • Constipation • Faecal incontinence • Neurogenic bowel **Sexual dysfunction** • Erectile/ejaculatory dysfunction • Altered sexual sensation Other autonomic abnormalities may include impaired sweating and temperature regulation --- **5 Respiratory Features** Respiratory impairment depends on the level of injury • Difficulty in deep inspiration • Weak cough • Retained respiratory secretions • Reduced vital capacity • Respiratory muscle paralysis • Respiratory failure in high cervical injuries Injury involving C3 C5 can compromise diaphragmatic function because the phrenic nerve arises from these cervical segments --- **6 Cardiovascular Features** Particularly with cervical and high thoracic injuries • Hypotension • Bradycardia • Peripheral vasodilation • Neurogenic shock • Cardiac rhythm disturbances Neurogenic shock results from loss of

sympathetic outflow and is particularly associated with injuries above approximately T6 --- **7 Clinical Patterns of Incomplete SCI**

- A Central Cord Syndrome** • Usually follows cervical hyperextension injury • Upper limbs are affected more than lower limbs • Hand weakness may be prominent • Sensory disturbance may occur
- B Brown Séquard Syndrome** Due to hemisection of the spinal cord • Ipsilateral motor weakness • Ipsilateral loss of proprioception/vibration • Contralateral loss of pain and temperature
- C Anterior Cord Syndrome** • Motor paralysis below the lesion • Loss of pain and temperature • Relative preservation of dorsal column functions such as vibration and proprioception --- **8 Complete and Incomplete Injury**
- Complete SCI** • No sensory or motor function is preserved in the sacral segments S4 S5 • No sacral sparing • Classified as AIS/ASIA Grade A
- Incomplete SCI** • Some sensory and/or motor function is preserved below the neurological level including sacral function • Classified as AIS B D depending on the amount of preserved function • AIS E normal motor and sensory examination --- **9 Clinical Examination**

Assessment should include • Level of neurological injury • Motor power • Sensory examination • Deep tendon reflexes • Anal sensation and voluntary anal contraction • Muscle tone • Respiratory function • Bladder and bowel function • Autonomic function

The International Standards for Neurological Classification of Spinal Cord Injury ISNCSCI/ASIA examination is used to determine the neurological level and severity of injury --- **10 Important Complications**

- Respiratory failure • Neurogenic shock • Spinal shock • Pressure sores • Deep vein thrombosis and pulmonary embolism • Urinary tract infection • Neurogenic bladder and bowel • Contractures • Spasticity • Autonomic dysreflexia • Chronic pain • Permanent paralysis